End Semester Exam: *CHM204: Electrochemistry, Colloids, and Interfaces*

Date (Day) / Time / Venue: 7-May-2026 (Thursday) / 2:30 PM to 5:30 PM / LH6-LHC

Part - B

Provide a brief and to-the-point answer to ALL questions. **Marks: 4 x 10 = 40**

Useful relations:

Lorentz-Lorentz equation:

$$\frac{n^2-1}{n^2+2}=\frac{N.\alpha}{3\varepsilon_0}$$

Physical constants:

$\varepsilon_0 = 8.854 \times 10^{-12}\, F.m^{-1}$

$\gamma_{H_2O} = 0.072\, N.m^{-1}$ at 20°C

$1D = 3.336 \times 10^{-30}\, C.m$

$R = 8.314\, J.K^{-1}.mole^{-1}$

1) Suppose an H_2O molecule (μ = 1.85 D) approaches an anion. What is the favorable orientation of the molecule? Calculate the electric field (in volts per meter) experienced by the anion when the water dipole is (a) 1.0 nm, (b) 0.3 nm, (c) 30 nm from the ion. The polarizability volume of H_2O at optical frequencies is 1.5×10^{-24} cm^3: estimate the refractive index of water. The experimental value is 1.33; what may be the origin of the discrepancy?

2 + 2 + 2 + 3 + 1 = 10

2) A solution consists of 25 percent by mass of a trimer with M = 22 kg mol^{-1} and its monomer. What average molar mass would be obtained from measurement of: (a) osmotic pressure, (b) light scattering? What is the role of cholesterol in the rigidity of the lipid bilayers? The radius of gyration of a long chain molecule is found to be 7.3 nm. The chain consists of C–C links. Assume the chain is randomly coiled and estimate the number of links in the chain. Give an example of a liquid dispersion inside a solid bulk medium. Discuss the physical origin of the depletion interaction between two colloidal particles.

2 + 2 + 1 + 2 + 1 + 2 = 10

3) Calculate the pressure differential of water across the surface of a spherical air bubble of radius 200 nm at 20°C. The contact angle for water on clean glass is close to zero. Calculate the surface tension of water at 30 °C given that at that temperature water climbs to a height of 9.11 cm in a clean glass capillary tube of internal radius 0.320 mm. The density of water at 30°C is 0.9956 g cm^{-3}. Why are dust particles needed for cloud formation? Schematically draw the potential energy profiles for the dissociative chemisorption of an A_2 molecule if the chemisorption is activated and if it is not activated. What is the main difference between the Langmuir–Hinshelwood and Eley–Rideal mechanisms for surface-catalyzed reactions?

1 + 2 + 2 + 4 + 1 = 10

4) What are the differences between the Helmholtz model and the Gouy–Chapman model of the structure of the double layer? For how long on average would an H atom remain on a surface at 300 K and at 1,000 K if its desorption activation energy was 15 kJ mol^{-1}? Take τ = 0.10 ps at 0°C. What is overpotential? Discuss the chemical origins of corrosion and useful strategies for preventing it.

1 + 6 + 1 + 2 = 10

ALL THE BEST!!!